

Event Registration and Ticket Management System

project report

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INTRODUCTION

Application Overview

The Event Management System is a desktop-based Java application designed to manage users, events, registrations, and notifications.

The application supports three main roles:

1. Admin
 - Logs into a dedicated admin panel
 - Can view and delete events
2. Organizer
 - Creates events
 - Views the list of their events
 - Views attendee lists for each event
3. Attendee
 - Browses available events
 - Registers for events
 - Cannot register if the event is full
 - Receives notifications upon successful registration

The system ensures secure access through user authentication, hashed passwords, and prevents duplicate registrations or invalid operations.

Database Connection

The system uses MySQL as the database and connects using JDBC through the DBConnection class.

The connection process includes:

- Loading the MySQL JDBC driver
- Creating a connection using URL, username, and password
- Handling SQL exceptions
- Ensuring safe closing of statements and result sets

Main tables:

- users
- admins
- events
- categories
- registrations
- notifications

Each table has proper foreign keys, unique constraints, and auto-increment primary keys.

GUI Elements

The application was built entirely using Java Swing, and uses the following GUI components:

1-Buttons (JButton),Used for actions such as:

- Login
- Register
- Create Event
- Delete Event
- View Attendees
- Navigate between windows

2-Text Fields (JTextField, JPasswordField)

Used to collect:

- Username
- Email
- Password
- Event information (name, location, seats)

3-Combo Boxes (JComboBox)

Used for selecting:

- Gender
- User role (Organizer / Attendee)
- Event category

4-Tables (JTable)

Used for:

- Displaying events
- Showing attendee lists
- Viewing admin panels

5-Panels (JPanel)

All GUI screens are arranged using:

- BorderLayout
- GridLayout
- BoxLayout

To achieve a clean, organized interface.

Exception Handling

4. Exception Handling

The application has extensive exception handling to prevent system crashes and ensure good user experience.

Examples of exceptions handled:

User Input Exceptions

- Empty username or password fields
- Invalid email format
- Missing gender or role selection

Authentication Exceptions

- Incorrect username or password
- Already registered users trying to register again
- Admin login failures

Event Exceptions

- User registering for a full event
- Attempt to create an event with invalid values
- Deleting an event with foreign key constraints

Database Exceptions

- SQL connection failures
- Query execution errors
- Duplicate key violations

All exceptions show clear, user-friendly messages using JOptionPane.

Testing and Validation

Testing was performed for all major features, including:

User Authentication

- Valid and invalid login attempts
- Registering new users
- Ensuring hashed passwords verify correctly

Organizer Features

- Creating events
- Displaying events
- Viewing attendee lists

Attendee Features

- Registering for events
- Preventing duplicate registration
- Blocking registration when event is full
- Receiving notification messages

Admin Features

- Logging in
- Deleting events
- Viewing all system information

Database Integrity Tests

- Foreign key enforcement
- Ensuring UNIQUE(event_id, attendee_id) works
- Notifications linked correctly to users and events

All features were validated manually through the GUI and MySQL queries.

TOOLS AND TECHNOLOGIES USED

1-Programming Language

Java : used to build the application logic and handle system operations.

2-Graphical User Interface

Java swing :Used to design all windows and screens such as:

- Login & Registration forms
- Admin dashboard
- Organizer dashboard
- Events panel
- Tickets panel
- Notifications panel
- Includes components like JButton, JTable, JPanel, JComboBox, JTextField, JPasswordField.

3-IDE (Development Environment)

NetBeans — used to write, organize, and run the project's Java code.

4-Database

MySQL

Used to store application data such as:

- Users & Admins
- Events
- Categories
- Registrations
- Notifications

All tables use primary keys, foreign keys, constraints, and timestamps.

5-Database Connectivity

JDBC (Java Database Connectivity): used to connect the Java application to the database and perform SQL operations (insert, update, delete, search).

TOOLS AND TECHNOLOGIES USED

6-Project File Management

Google Drive / OneDrive:

Used to save the project files and keep updated versions, and help the team share and access the latest work easily

7-Exception Handling Tools

Java try/catch + JOptionPane

Used to handle:

- Login errors
- Invalid inputs
- Full-event registrations
- Database connection failures
- SQL execution errors

8- Libraries / Components

Java Swing components used in the GUI:

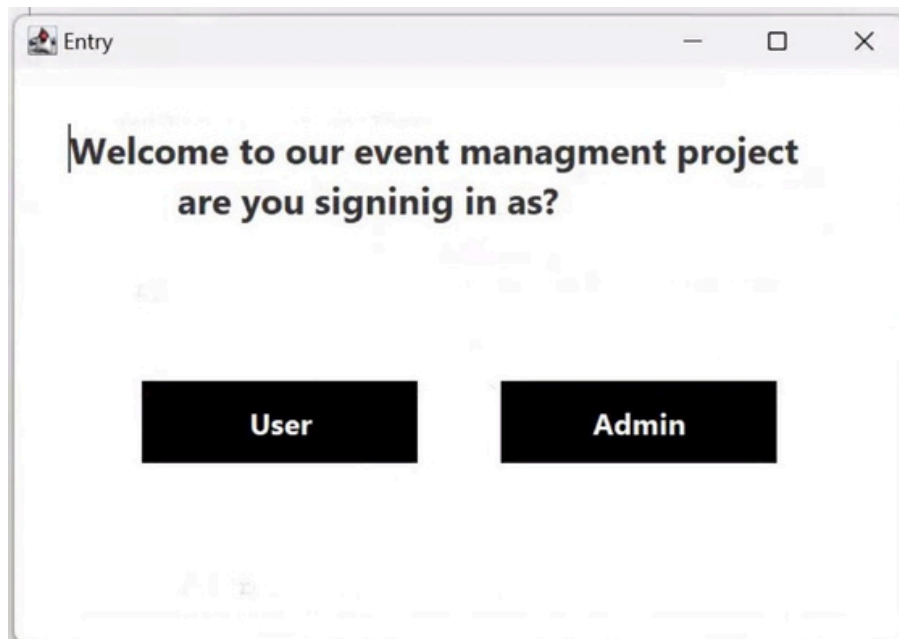
- JTable
- JOptionPane
- JComboBox
- JTextField
- JPasswordField
- Other layout managers like BorderLayout, GridLayout, BoxLayout

9-Password Hashing (Security Tool)

Password Hashing Utilities

Used to convert user passwords into a secure format before storing them in the database to protect user information.

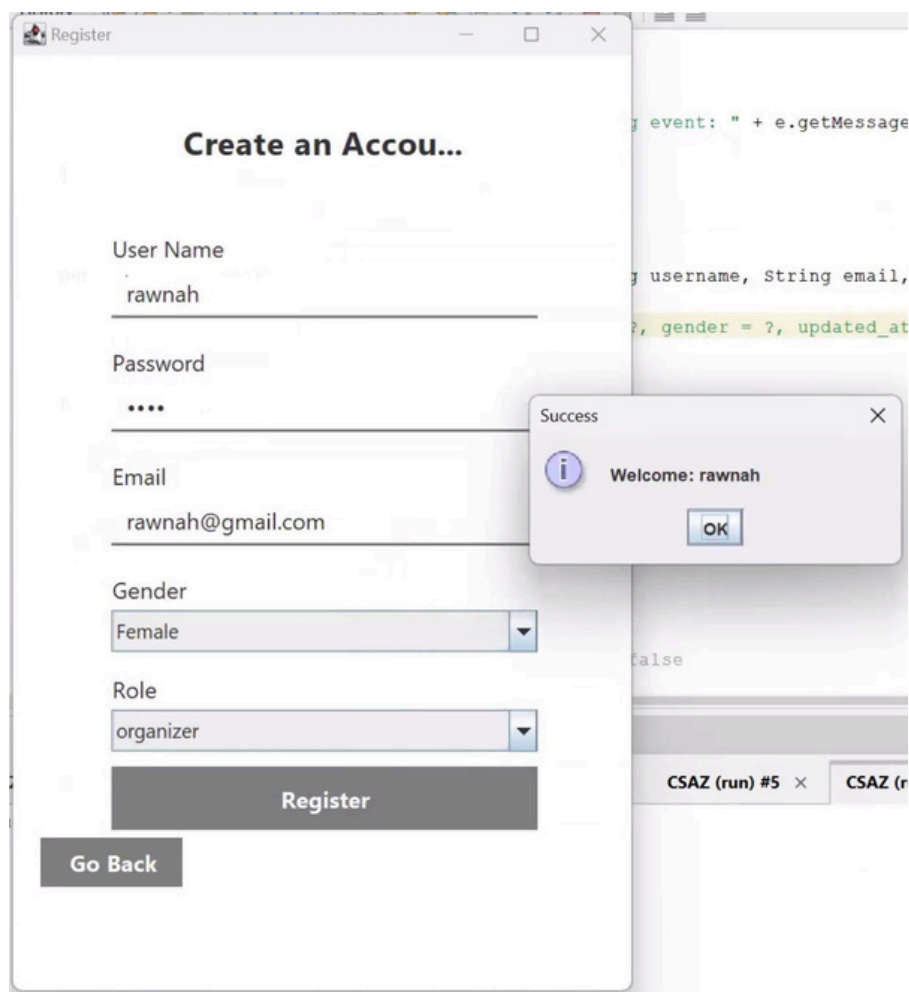
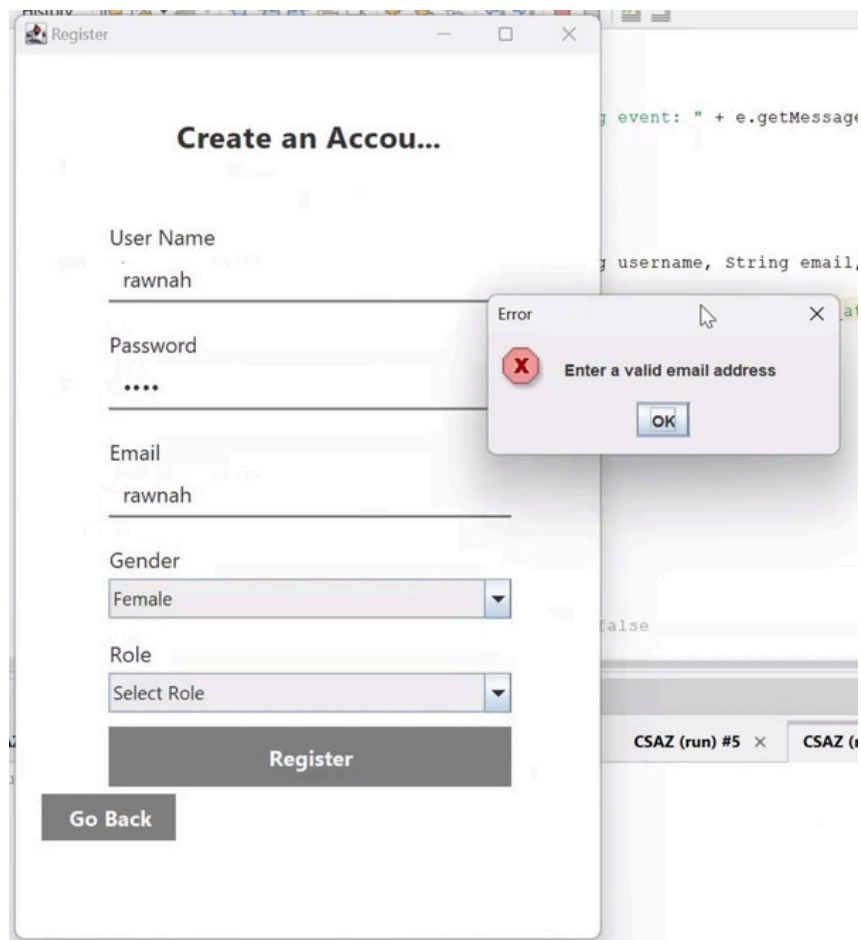
OUTPUT :

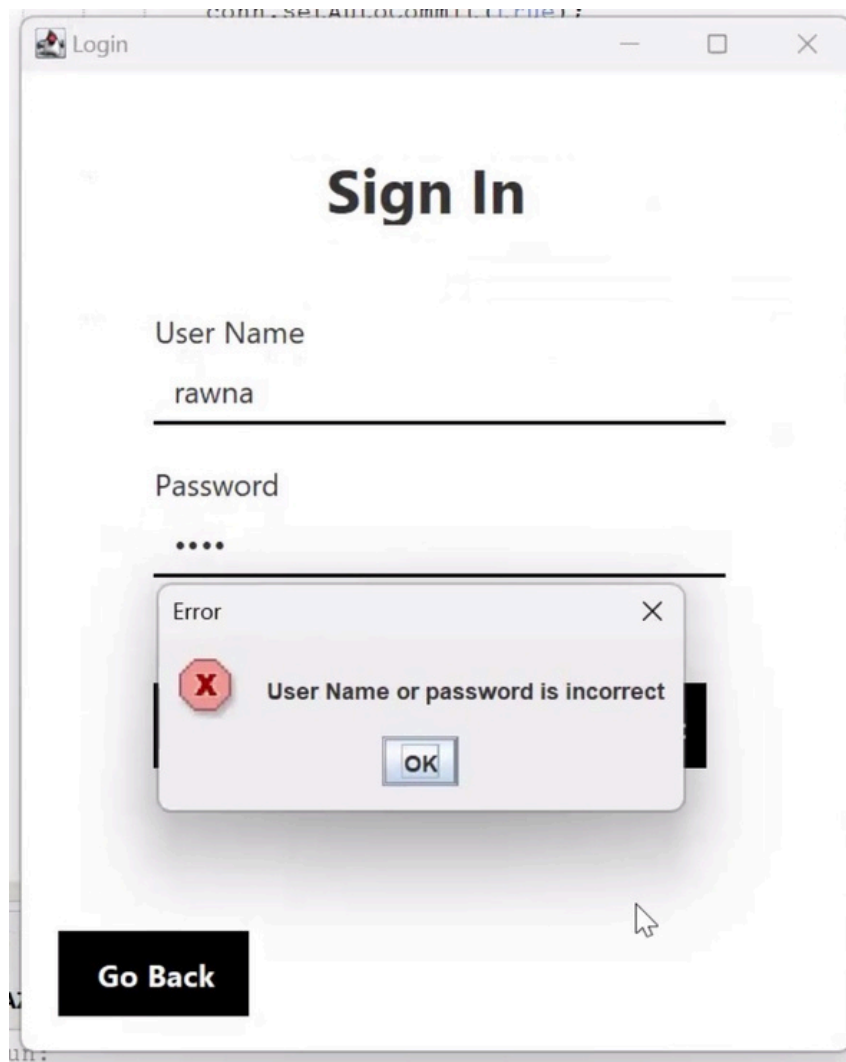


USER INTERFACE :

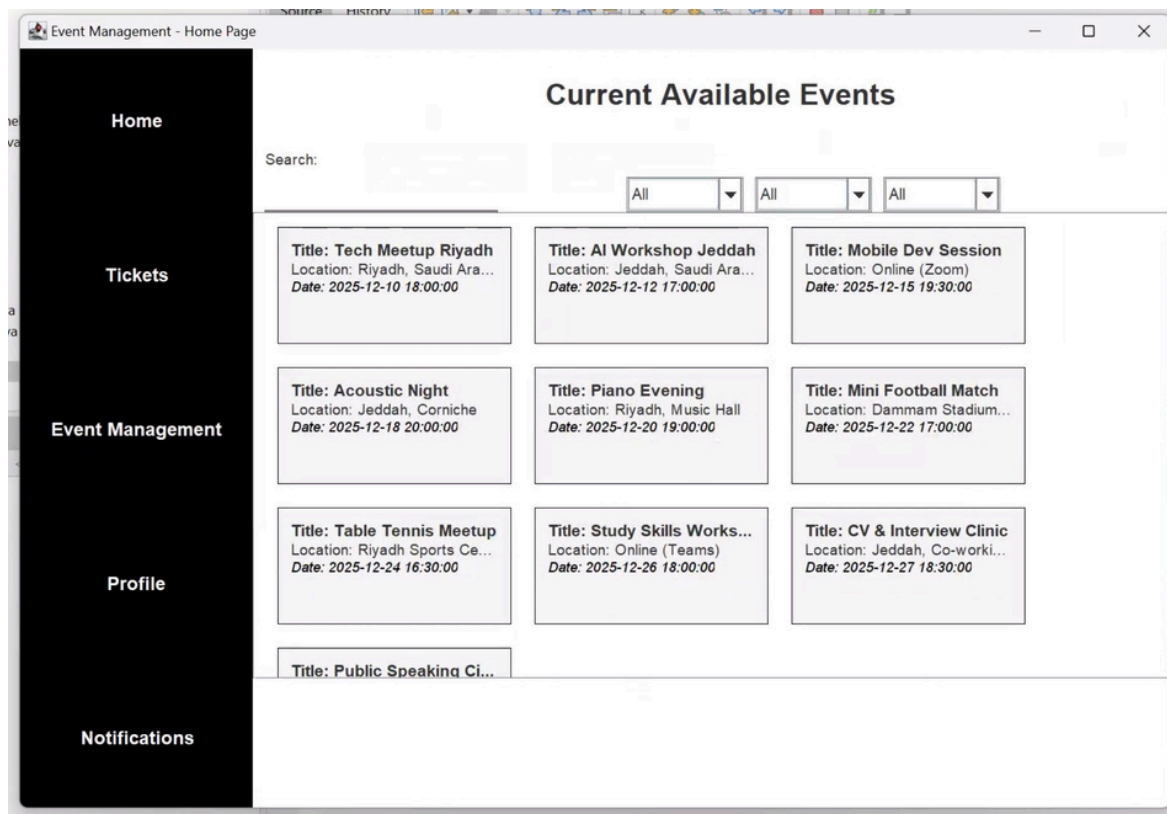
The 'Login' window is titled "Sign In". It contains two input fields: "User Name" and "Password". Below these fields are two black buttons with white text: "Login" and "Create Account". At the bottom left, there is a black button with white text: "Go Back".

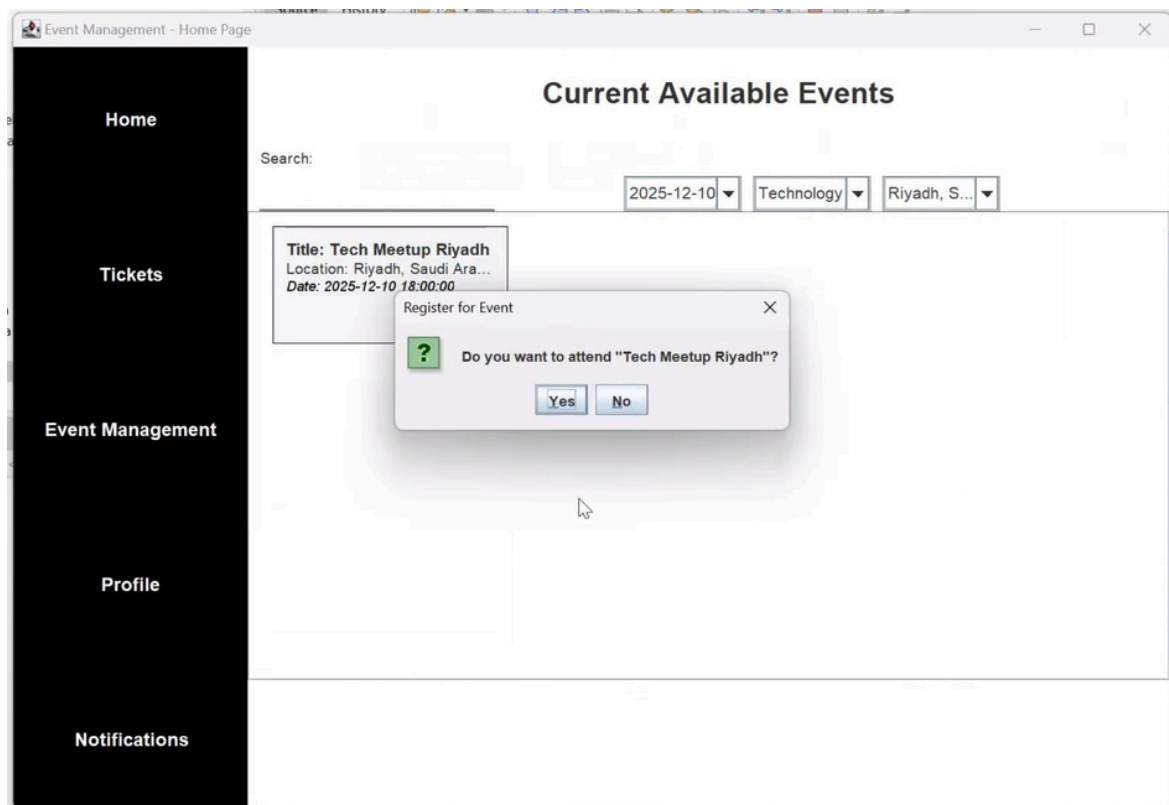
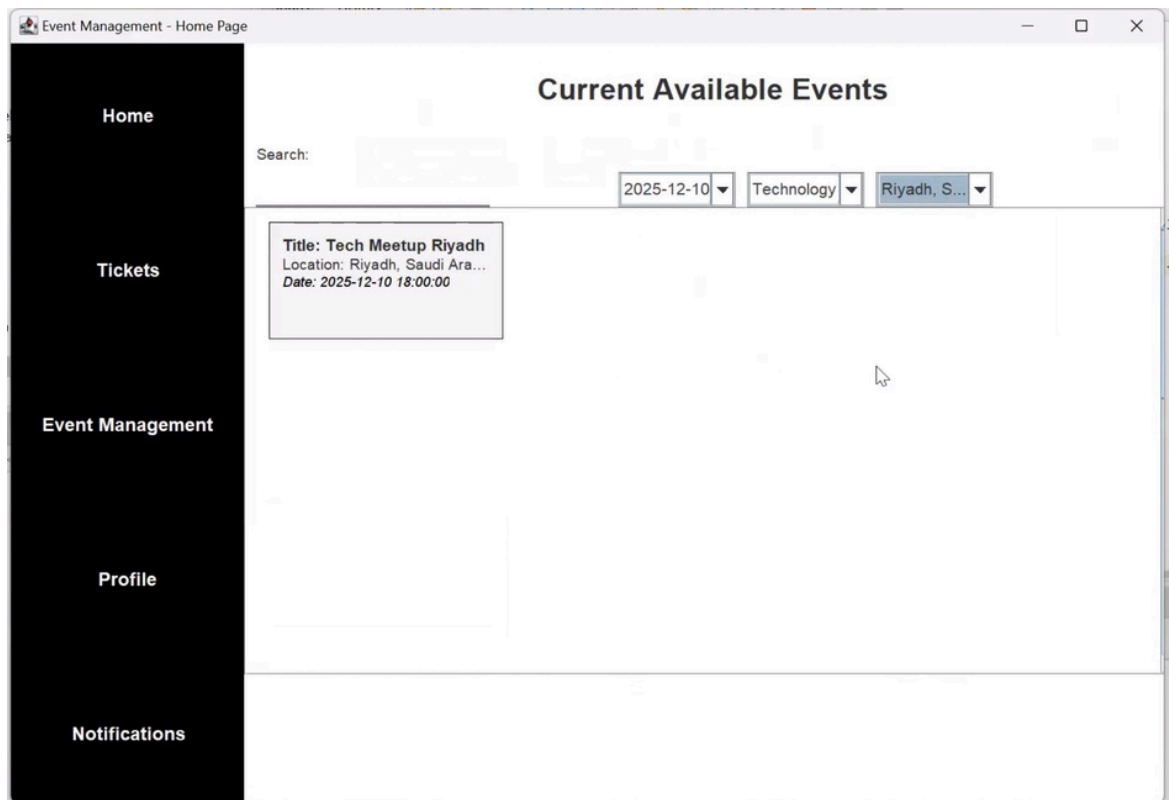
The 'Register' window is titled "Create an Accou...". It contains four input fields: "User Name", "Password", "Email", and "Gender" (a dropdown menu with "Select Gender" as the placeholder). Below these fields is a "Role" dropdown menu with "Select Role" as the placeholder. At the bottom, there are two buttons: a black button with white text "Go Back" and a grey button with black text "Register".

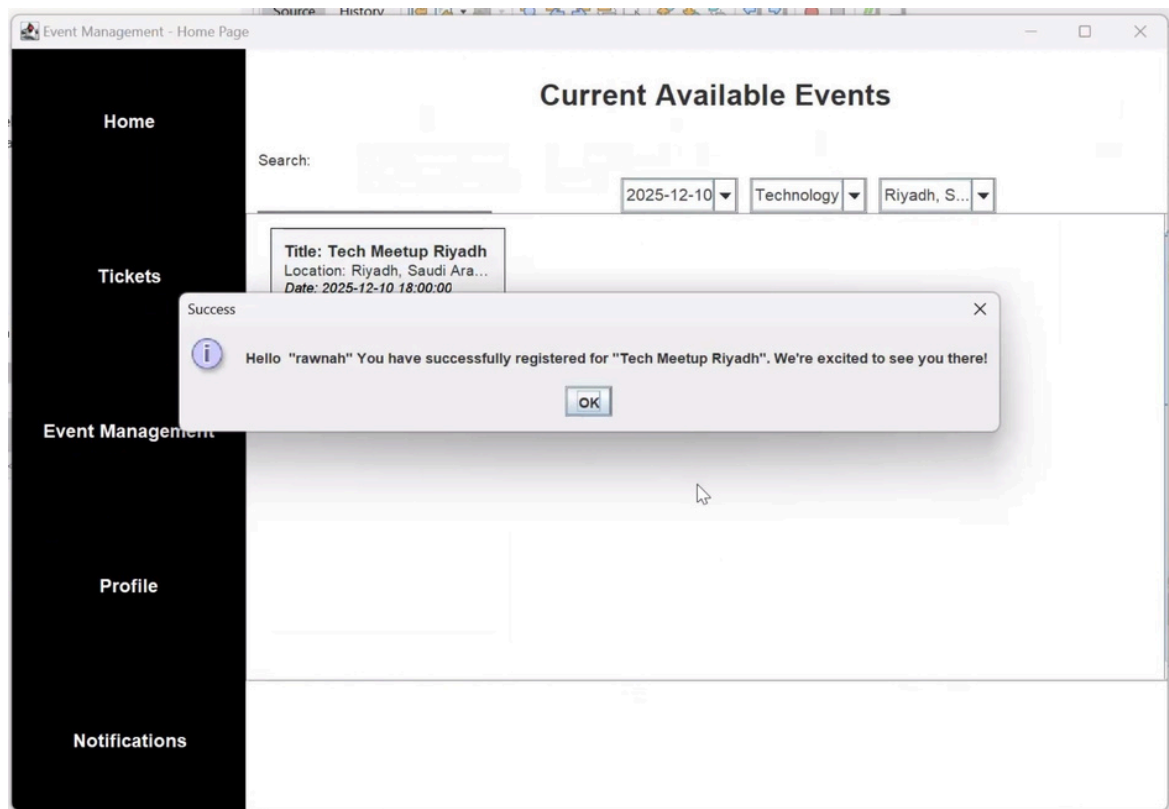




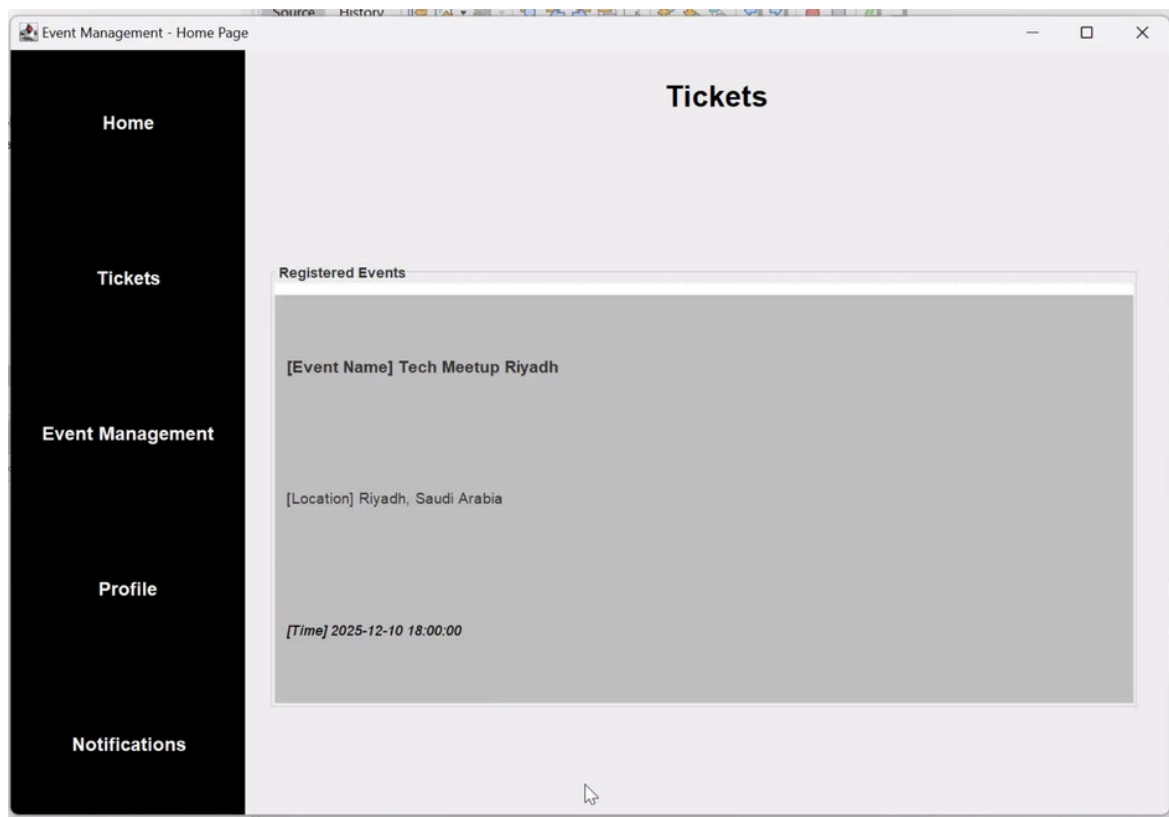
HOME :



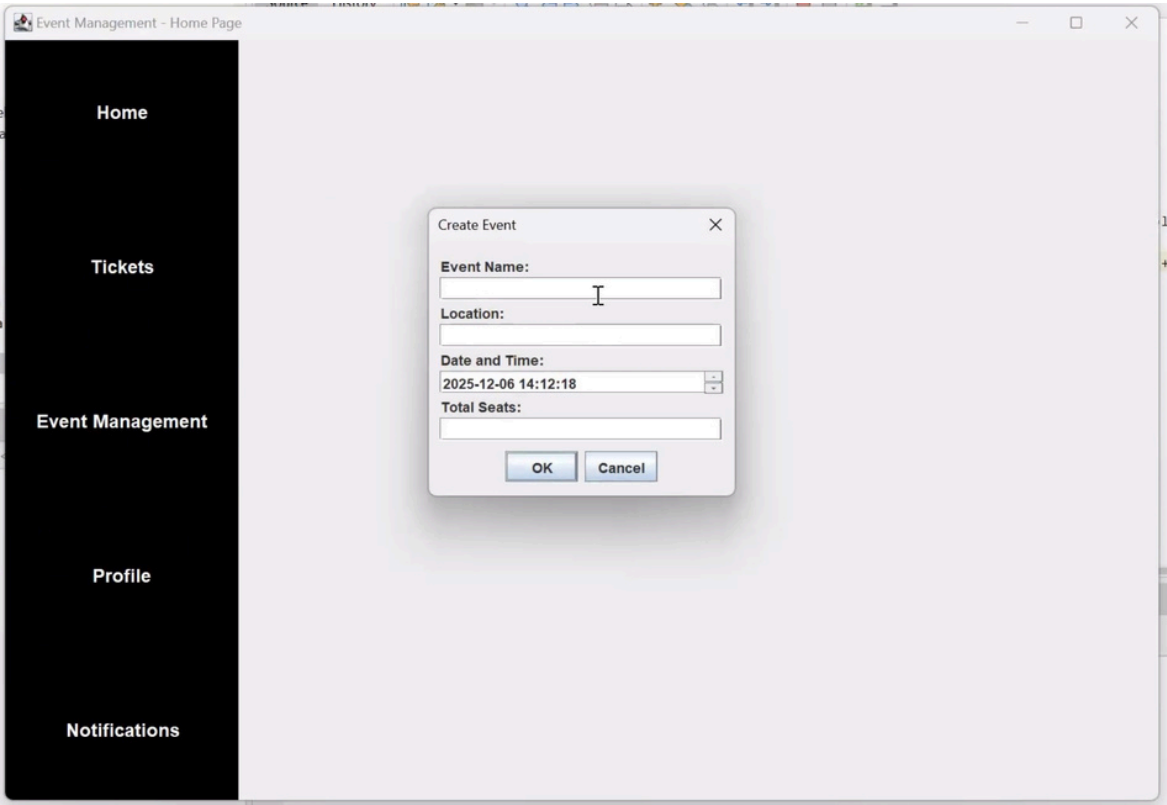
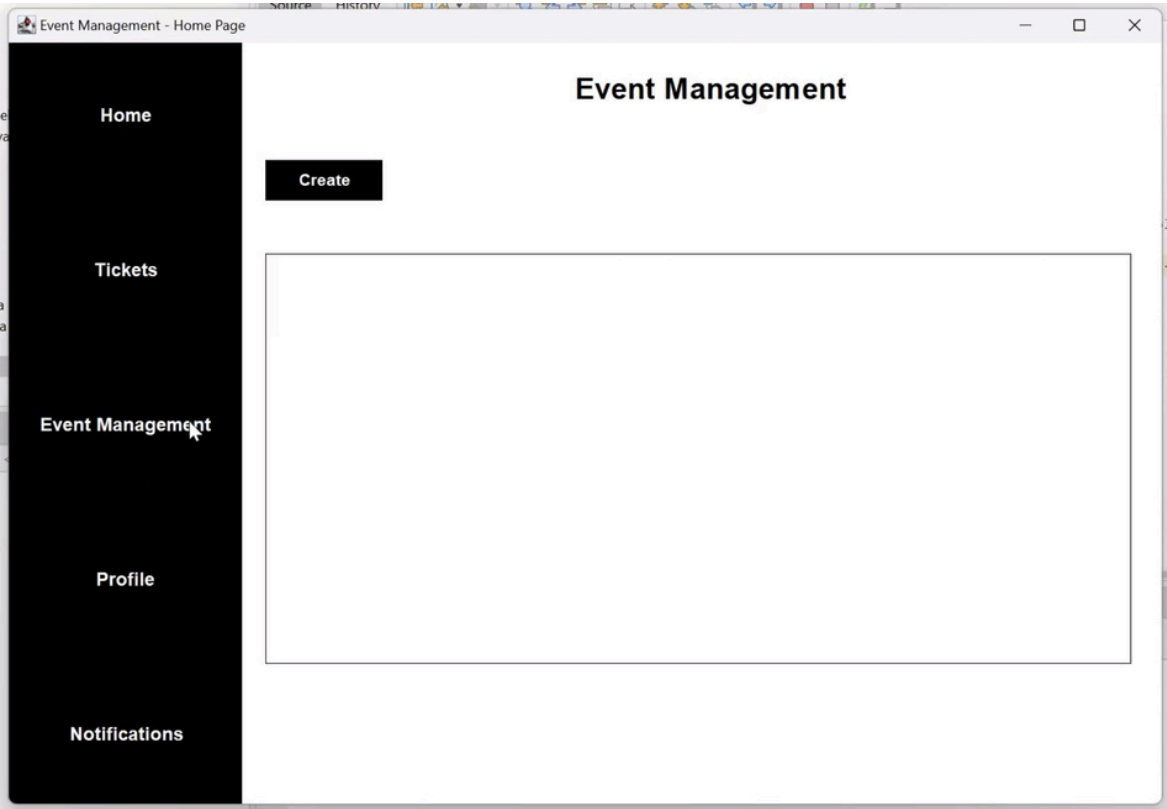


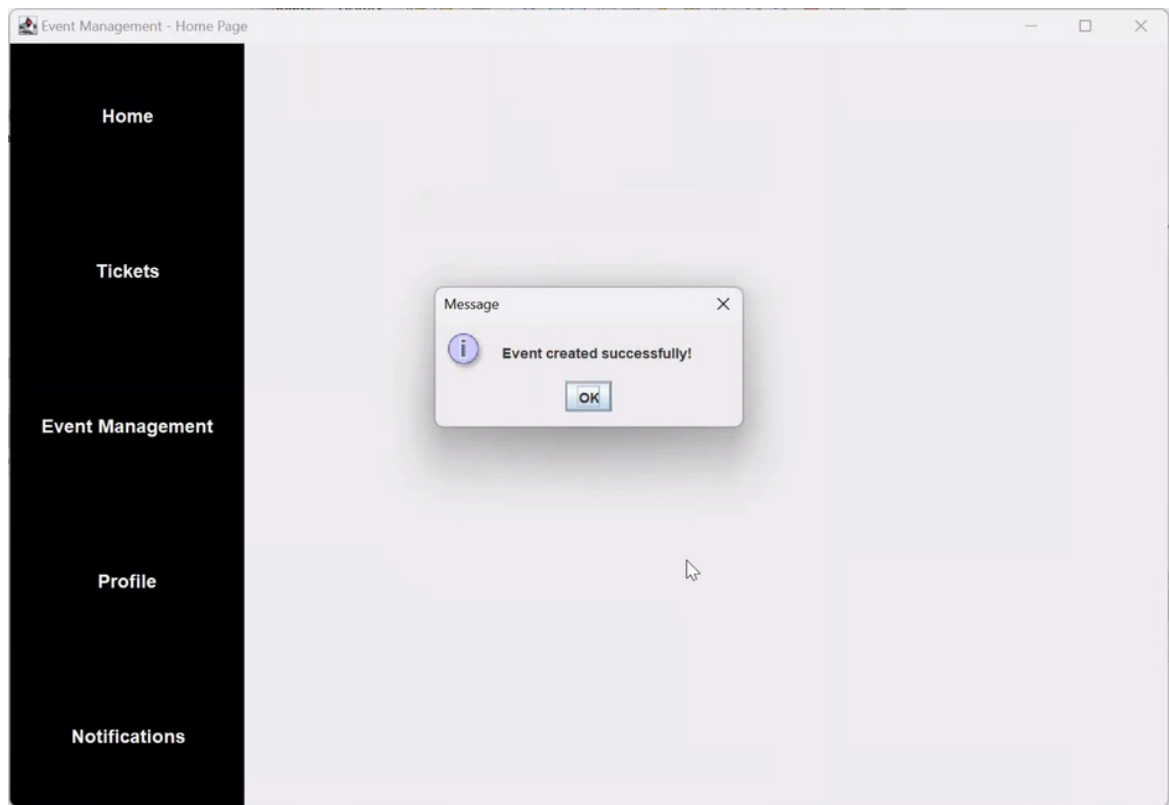


TICKETS :

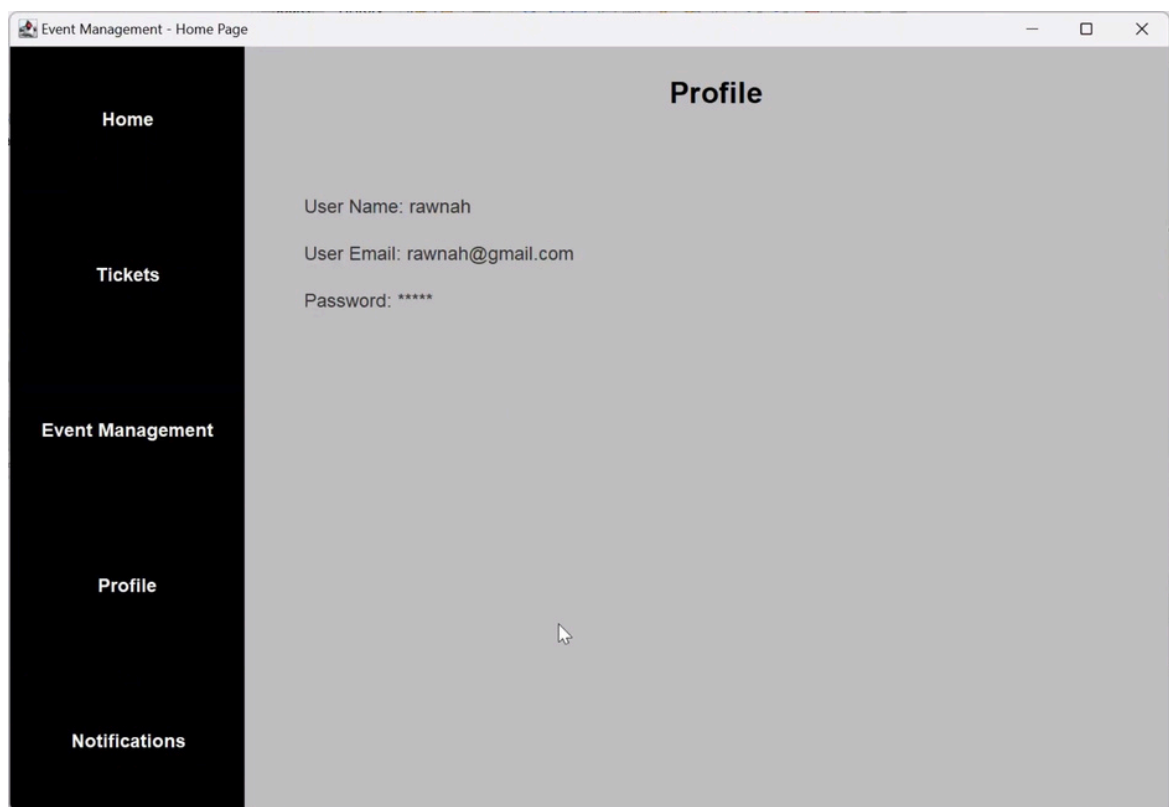


EVENT MANAGMENT :

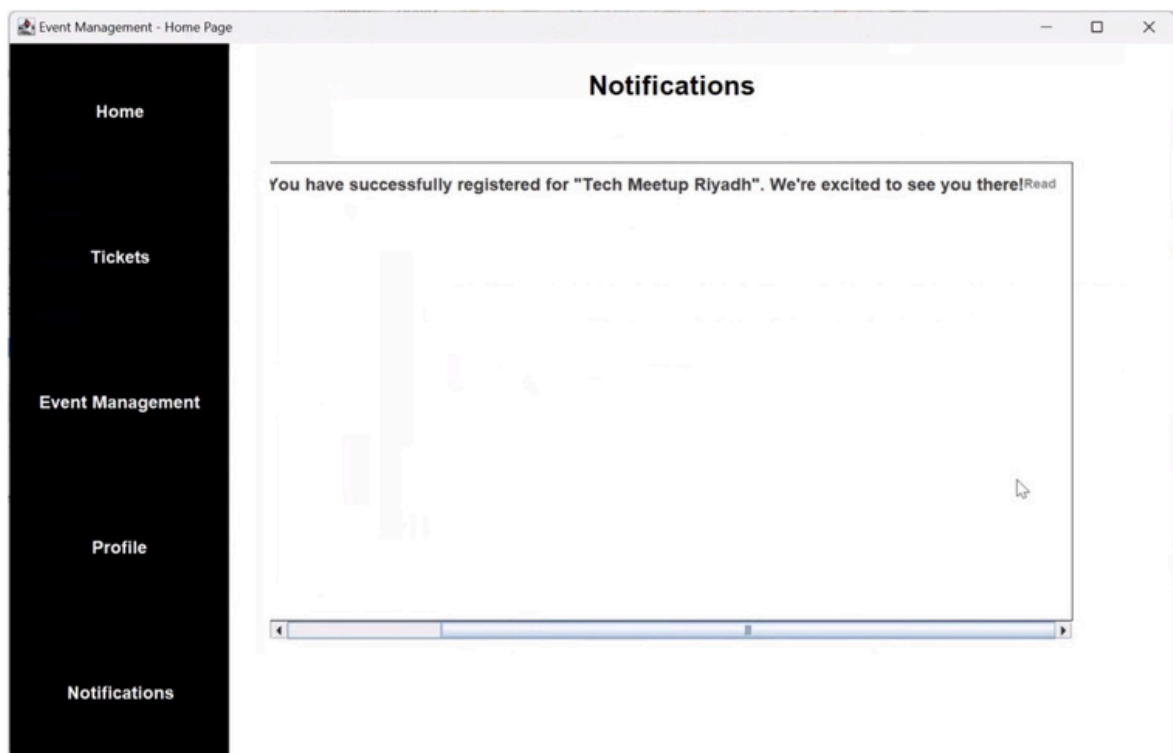
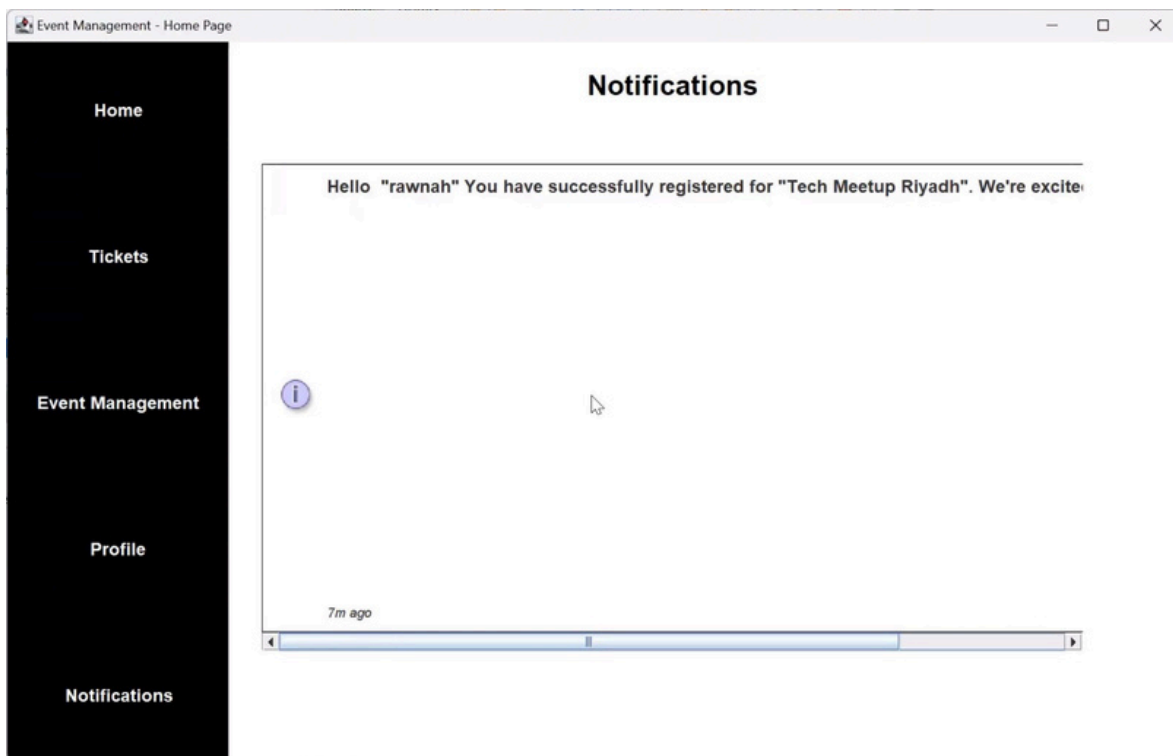




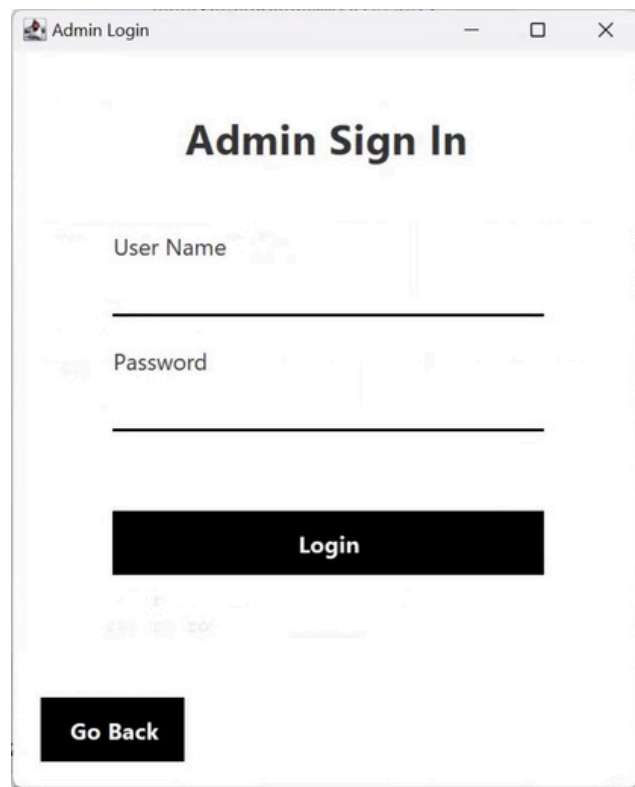
PROFILE :



NOTIFICATIONS :



ADMIN INTERFACE :



Admin Login

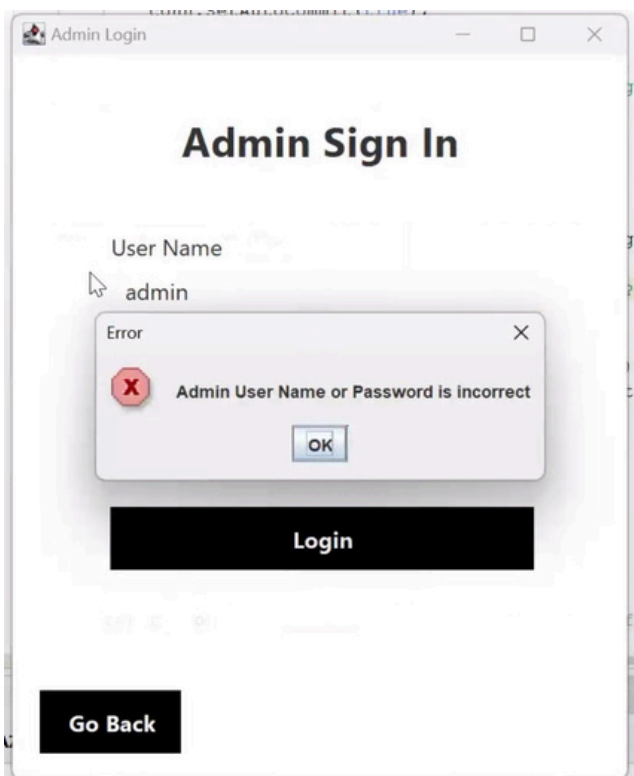
Admin Sign In

User Name

Password

Login

Go Back



Admin Login

Admin Sign In

User Name

admin

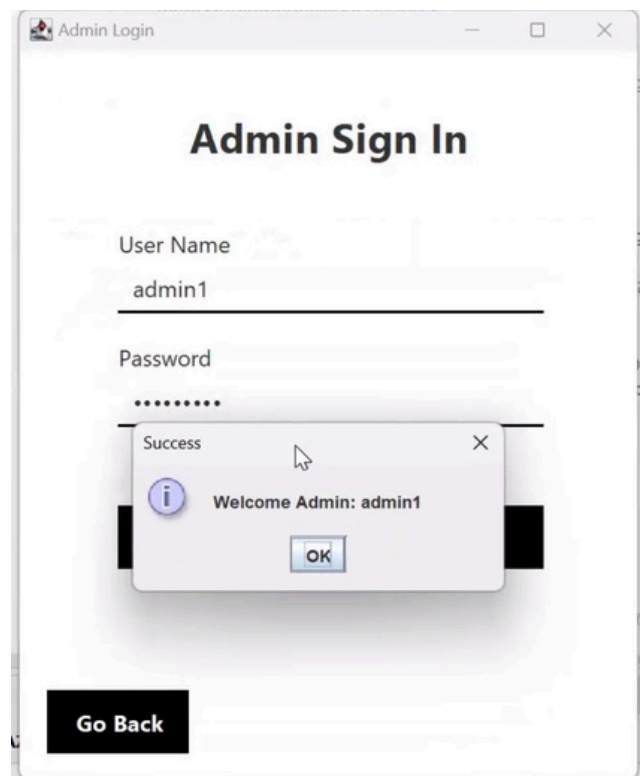
Error

Admin User Name or Password is incorrect

OK

Login

Go Back



Admin Login

Admin Sign In

User Name

admin1

Password

.....

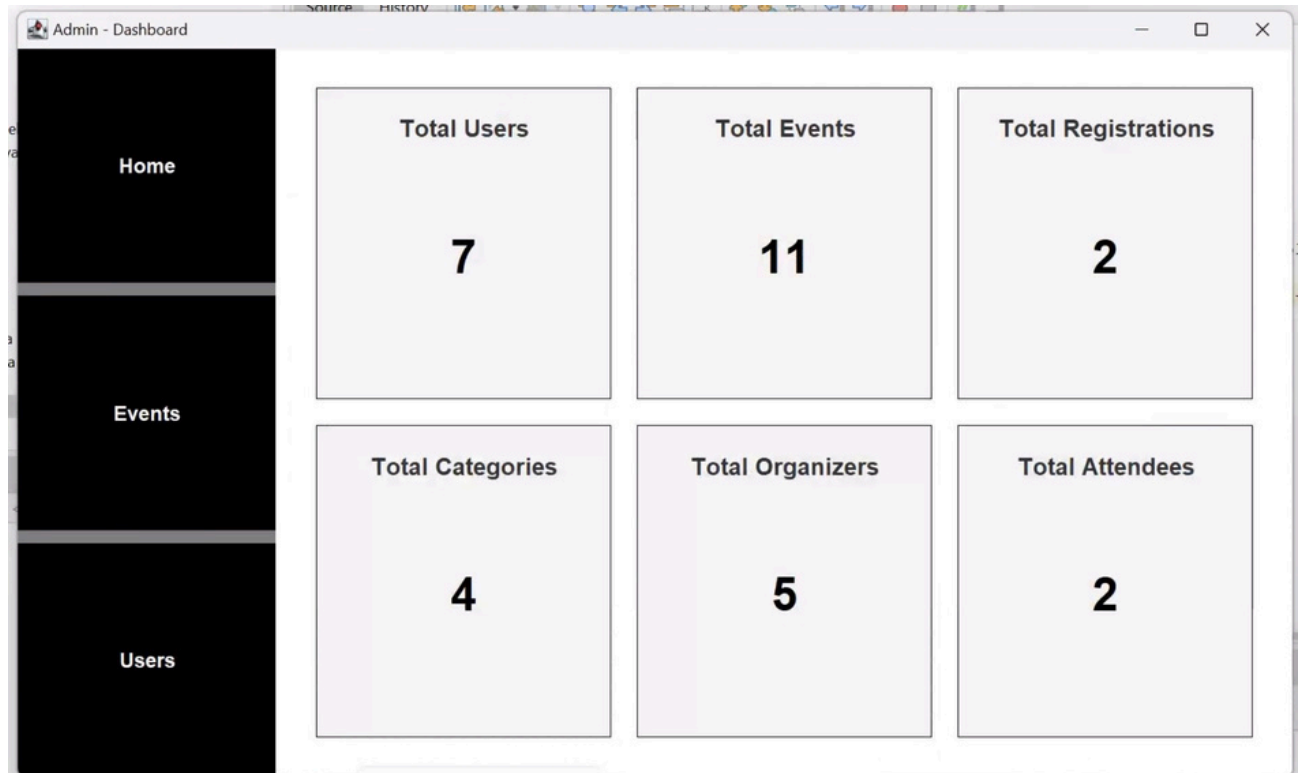
Success

Welcome Admin: admin1

OK

Go Back

HOME :



EVENTS MANAGEMENT :

The Admin Dashboard Events Management view displays a table of events. The table has the following columns: Name, Date, Location, Description, Total Seats, Registrati..., Most atte..., Availability, and Action. The table contains 8 rows of event data, each with a 'Delete' button in the Action column.

Name	Date	Location	Description	Total Seats	Registrati...	Most atte...	Availability	Action
Tech Meet...	2025-12-1...	Riyadh, Sa...	Small tech networking event.	3	2	Yes	Yes	Delete
AI Worksh...	2025-12-1...	Jeddah, S...	Hands-on AI mini workshop.	2	0	No	Yes	Delete
Mobile De...	2025-12-1...	Online (Zo...	Intro to mobile development	3	0	No	Yes	Delete
Acoustic Ni...	2025-12-1...	Jeddah, C...	Small live acoustic music show.	3	0	No	Yes	Delete
Piano Eve...	2025-12-2...	Riyadh, Mu...	Piano performance for limited audience.	2	0	No	Yes	Delete
Mini Footb...	2025-12-2...	Dammam ...	Friendly small football match.	3	0	No	Yes	Delete
Table Tenn...	2025-12-2...	Riyadh Sp...	Casual table tennis gathering.	2	0	No	Yes	Delete
			Workshop on study and					

Admin - Dashboard

Events Management

Name	Date	Location	Description	Total Seats	Registrati...	Most atte...	Availability	Action
			music show.					
Piano Eve...	2025-12-2...	Riyadh, Mu...	Piano performance for limited audience.	2	0	No	Yes	Delete
Mini Footb...	2025-12-2...		Friendly small football.				Yes	Delete
Table Tenn...	2025-12-2...						Yes	Delete
Study Skill...	2025-12-2...	Online (Te...	on study and focus skills.	3	0	No	Yes	Delete
CV & Inter...	2025-12-2...	Jeddah, C...	Career clinic for students.	3	0	No	Yes	Delete
Public Spe...	2025-12-2...	Riyadh, M...	Practice public speaking in small circle.	3	0	No	Yes	Delete
Winter Wo...	2025-12-0...	Riyadh, Sa...		5000	0	No	Yes	Delete

?

Confirm Deletion

Are you sure you want to delete the event "Winter Wonderland"?

Yes No

Admin - Dashboard

Events Management

Name	Date	Location	Description	Total Seats	Registrati...	Most atte...	Availability	Action
			development					
Acoustic Ni...	2025-12-1...	Jeddah, C...	Small live acoustic music show.	3	0	No	Yes	Delete
Piano Eve...	2025-12-2...	Riyadh, Mu...	Piano performance			No	Yes	Delete
Mini Footb...	2025-12-2...	Dammam ...				No	Yes	Delete
Table Tenn...	2025-12-2...	Riyadh Sp...	tennis gathering.	2	0	No	Yes	Delete
Study Skill...	2025-12-2...	Online (Te...	Workshop on study and focus skills.	3	0	No	Yes	Delete
CV & Inter...	2025-12-2...	Jeddah, C...	Career clinic for students.	3	0	No	Yes	Delete
Public Spe...	2025-12-2...	Riyadh, M...	Practice public speaking in small circle.	3	0	No	Yes	Delete

i

Event deleted.

OK

USERS :

Admin - Dashboard

Users

User ID	Username	Email	Role	Gender	Created At	Updated At
1	rawan	rawan@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:45
2	aljazi	aljazi@example.com	organizer	Female	2025-12-05 22:10	
3	mohammed	mohammed@examp...	organizer	Male	2025-12-05 22:10	2025-12-06 13:57
4	rifa	rifa@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:49
5	toto	toto@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:25
7	rawnah	rawnah@gmail.com	attendee	Female	2025-12-06 14:08	2025-12-06 14:49

Edit Selected User Delete Selected User

Admin - Dashboard

Users

User ID	Username	Email	Role	Gender	Created At	Updated At
1	rawan	rawan@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:45
2	aljazi	aljazi@example.com	organizer	Female	2025-12-05 22:10	
3	mohammed	mohammed@examp...	organizer	Male	2025-12-05 22:10	2025-12-06 13:57
4	rifa	rifa@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:49
5	toto	toto@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:25
7	rawnah	rawnah@gmail.com	attendee	Female	2025-12-06 14:08	2025-12-06 14:49

Edit User

Username: aljazi

Email: aljazi@example.com

Role: attendee

Gender: Female

OK Cancel

Edit Selected User Delete Selected User

Admin - Dashboard

Users

User ID	Username	Email	Role	Gender	Created At	Updated At
1	rawan	rawan@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:45
2	aljazi	aljazi@example.com	organizer	Female	2025-12-05 22:10	
3	mohammed	mohammed@examp...	organizer	Male	2025-12-05 22:10	2025-12-06 13:57
4	rifa	rifa@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:49
5	toto	toto@example.com	organizer	Female	2025-12-05 22:10	2025-12-06 14:25
7	rawnah	rawnah@gmail.com	attendee	Female	2025-12-06 14:08	2025-12-06 14:49

Success

User updated successfully.

OK

Edit Selected User Delete Selected User

